機械力学用語

2022年4月17日　　13:28

1,常微分方程式の解法

1, 仮定　$x = Ae^{\lambda t}$

2, 特性方程式によって，　特性方程式の根(λ)を求める

①不重根：$x = x_1 + x_2 = A_1 e^{\lambda_1 t} + A_2 e^{\lambda_2 t}$

②重根：$x = A_1 e^{\lambda t} + A_2 \lambda e^{\lambda t}$

③複素数根：実部と虚部をわけて、オイラーの方程式を利用する（解の中での虚数を避けるため）

$$\lambda = a \pm ib$$
$$x = x_1 + x_2 = A_1 e^{\lambda_1 t} + A_2 e^{\lambda_2 t} = A_1 e^{(a+ib)t} + A_2 e^{(a-ib)t}$$
$$= e^a\left(A_1 e^{(ib)t} + A_2 e^{(-ib)t}\right)$$
$$= e^a\left(C_1 \cos bt + C_2 \sin bt\right)$$

$$(C_1 = A_1 + A_2, C_2 = i\left(A_1 - A_2\right))$$

3, 初期条件によって積分定数を決める.

$$\mathrm{x}(0) = x_0$$
$$\dot{x}(0) = v_0$$

2, 1自由系の自由振動

微分方程式の特解（まとめ）

$\ddot{x} + 2\varsigma\omega_n\dot{x} + \omega_n^2 x = \sin\omega t$ の特解は以下の3パターン

i) $\varsigma > 0$ のとき

$x(t) = A\sin\omega t + B\cos\omega t$ と仮定して代入

ii) $\varsigma = 0$ であり $\omega_n \neq \omega$ のとき

$x(t) = A\sin\omega t$ と仮定して代入

iii) $\varsigma = 0$ であり $\omega_n = \omega$ のとき

$x(t) = At\cos\omega t$ と仮定して代入

※上記のiii)では，外力が固有角振動数に一致して解が発散する.

3, 1自由系の非減衰強制振動と減衰強制振動

1自由度系の強制振動

制御工学の知識が使いやすい

伝達率:

4, 2自由度の運動方程式

運動方程式の導出:

$$\mathbf{M}\ddot{\boldsymbol{x}} + \boldsymbol{C}\dot{\boldsymbol{x}} + \boldsymbol{K}\boldsymbol{x} = \boldsymbol{F}$$

5, 2 自由度の非減衰自由振動の解

1,運動方程式の導出

2,運動方程式の解を仮定して代入する.

$$x_1 = \boldsymbol{u_1} \cos \boldsymbol{\omega t}\ , x_2 = \boldsymbol{u_2} \cos \boldsymbol{\omega t}$$

3,振動数方程式の導出(特性方程式)

$\omega < 0$の解は無意味な解である

$\omega_1 < \omega_2 < \cdots < \omega_n,$

1次固有振動数,2次固有振動数,⋯ n次固有振動数

固有振動モード(固有ベクトル)の導出(試験に重要)

固有ベクトルを求める

初期変位は固有モードの比と等しいと，振動幅はこの比率と等しい．

2自由度系非減衰強制振動

逆行列がある

反共振点$\omega = \omega_n = \sqrt{\frac{k}{m}}$

2自由度系の減衰強制振動

代入法(無理)

重要！伝達関数!

伝達率の比較

変位の応答倍率

加速度の応答倍率

動吸振器

C:

k:

ばね定数k2によって横軸方向に移動させる

最適調整手順：

実際の場合に，制御対象が一次自由系ではない方が多いので，これらの式は一切使えない

ラグランジュの方法

複雑な多自由度の運動方程式を「エネルギー」で求める.

一般化座標

座標変換(反対行列)

1回目　常微分方程式の解法

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2回目　1自由系の自由振動

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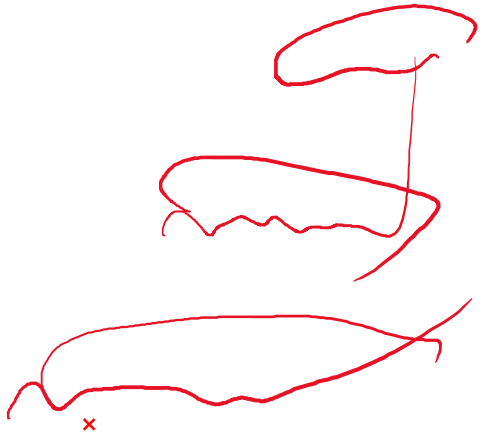

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3回目 1自由系の非減衰強制振動

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4回目 2自由度振動の運動方程式（試験範囲）

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5回目 2自由度非減衰自由振動(固有振動数)(試験に重要)

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(3) $m\ddot{x}_1 = k(x_2 - x_1)$

$2m\ddot{x}_2 = -k(x_2 - x_1)$

$$\begin{cases} m\ddot{x}_1 + kx_1 - kx_2 = 0 \\ 2m\ddot{x}_2 + kx_2 - kx_1 = 0 \end{cases}$$

$$\begin{bmatrix} m & 0 \\ 0 & 2m \end{bmatrix}\begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} k & -k \\ -k & k \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$x_1 = u_1 \cos\omega t$, $x_2 = u_2 \cos\omega t$ ( )

$$\begin{bmatrix} m & 0 \\ 0 & 2m \end{bmatrix}\begin{bmatrix} -u_1\omega^2\cos\omega t \\ -u_2\omega^2\cos\omega t \end{bmatrix} + \begin{bmatrix} k & -k \\ -k & k \end{bmatrix}\begin{bmatrix} u_1\cos\omega t \\ u_2\cos\omega t \end{bmatrix} = 0$$

$$\begin{bmatrix} -m\omega^2 + k & -k \\ -k & -2m\omega^2 + k \end{bmatrix}\begin{bmatrix} u_1\cos\omega t \\ u_2\cos\omega t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$m^2\omega^4 - 3km\omega^2 + k^2 - k^2 = 0$$

$$m\omega^2(m\omega^2 - 3k) = 0$$

$$\omega = 0 \quad , \quad \omega = \sqrt{\frac{3k}{m}}$$

$$\omega_1 = 0 \quad \omega_2 = \sqrt{\frac{3k}{m}}$$

6回目 固有振動モード(固有ベクトル)の導出

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7回目　2自由度系非減衰強制振動

2022年6月3日　8:50

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8回目 2自由度系の減衰強制振動

2022年6月8日　　12:21

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9回目 減衰のある動吸振器

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10回目 ラグランジュの方法による運動方程式の導出

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11回目 ラグランジュの方法の応用

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12回目　座標変換・モード座標による運動方程式

2022年6月29日　　23:11

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13回目　過去問演習

2022年7月6日　　19:09

14回目　複素解法(試験外)

2022年7月6日　　19:10

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15回目

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